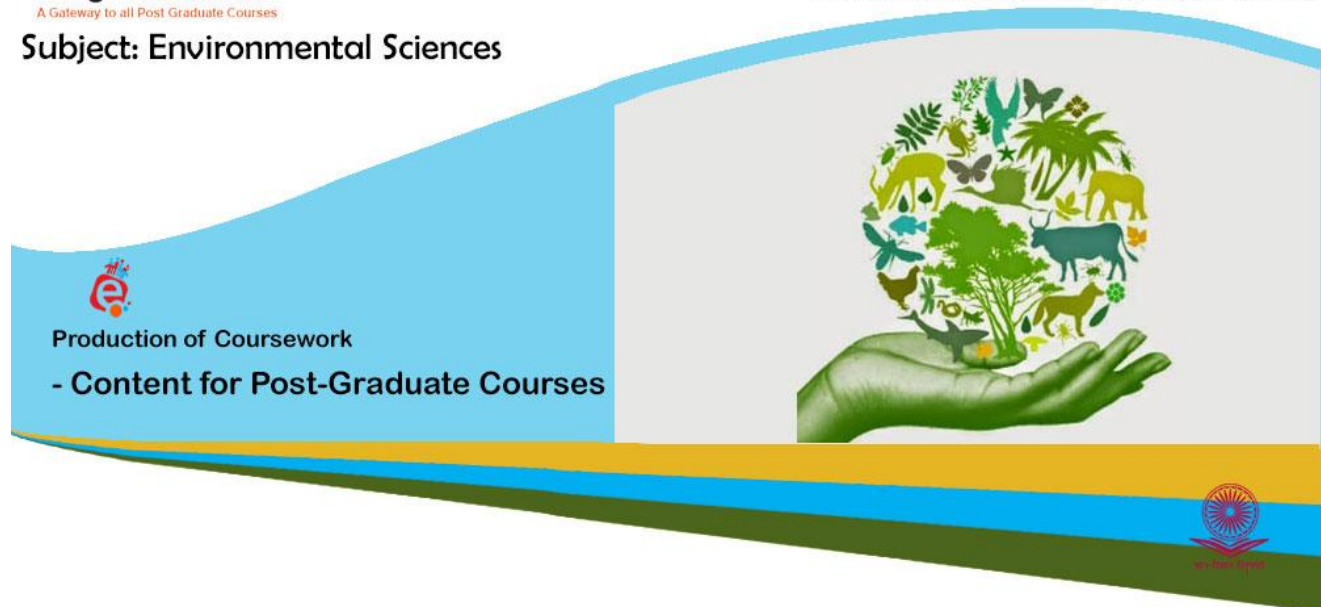


Subject: Environmental Sciences



Paper No: 13 Environmental Law and Policies

Module: 30 Wetland Protection



Development Team

Principal Investigator & Co- Principal Investigator	Prof. R.K. Kohli Prof. V.K. Garg & Prof. Ashok Dhawan Central University of Punjab, Bathinda
Paper Coordinator	Prof. P.S. Jaswal, Rajiv Gandhi National University of Law, Patiala
Content Writer	Dr. Renuka Soni Rajiv Gandhi National University of Law, Patiala
Content Reviewer	Prof. P.S. Jaswal, Rajiv Gandhi National University of Law, Patiala
Anchor Institute	 Central University of Punjab, Bathinda

Description of Module	
Subject Name	Environmental Sciences
Paper Name	Environmental Law and Policies
Module Name/Title	Wetland Protection
Module Id	EVS/ELP-XIII/30
Pre-requisites	Information about environmental protection .
Objectives	• To introduce the Concept of wetlands • To highlight the importance of Wetland • To Understand the legislative measures for protection of Wetlands • To highlight the various steps for the protection, conservation and management of wetlands
Keywords	Wetland Protection, Ramsar Convention, legislation,

WETLAND PROTECTION

LEARNING OBJECTIVES:

- To introduce the Concept of wetlands
- To highlight the importance of Wetland
- To Understand the legislative measures for protection of Wetlands
- To highlight the various steps for the protection, conservation and management of wetlands

1. INTRODUCTION

A wetland is a land area that is saturated with water, either permanently or seasonally in such a manner that acquires the features of a distinct ecosystem. However, what sets apart the wetlands from other forms like land, water bodies etc is the distinct vegetation of plants.

Wetlands are also considered the most biologically diverse of all ecosystems, serving as home to a wide range of plant and animal life, wetlands have played an important role in the protection of environment, however over the years wide spread exploitation of the earthly resources, pollution, industrialization, increasing human exploitation has gradually lead to wide spread exploitation of Wetlands. The exploitation and destruction of wetlands is a global phenomenon and is not confined to a particular country. However, gradually the issue of wetland protection has caught the attention of the policy makers across the globe. Before discussing the various aspects of Wetland Protection, it is imperative to under the concept of wetland.

2. DEFINITIONS

Though there are numerous acceptable definitions of Wetland. However, for the purpose of understanding, the meaning of the term 'Wetland', the same can be understood from a general perspective and standard acceptable definition as is available in various regulatory and policy decisions with regards to Wetlands.

a. General Definition

The term Wetland can be understood by distinguishing it from other forms of land and water bodies. Huge accumulation of water after a rain storm though makes the land wet, but the same

cannot be considered by any stretch of imagination as a wetland, infact, wetlands are generally characterized as having a water table that is near to a land for a long duration each year to support aquatic plants. A more precise definition of Wetland is that Wetland is a community, which is composed of Hybrid soil and Hydrophytes.

b. Definition according to Ramsar Convention, 1971

Ramsar Convention on Wetlands was signed in 1971 and the Convention is named after the City of Ramsar in Iran. Under the Ramsar Convention the term Wetland is defined in Article 1.1 and 2.1 of the Convention. According to Article 1.1 of Ramsar Convention, Wetlands are areas of marsh, fen, peat-land or water whether natural or artificial, permanent or temporary, with water i.e. static or flowing, fresh, brackish or salt, including areas of marine water, the depth of which at low tide does not exceed six meters.

According to Article 2.1, Wetland may incorporate riparian and coastal zones adjacent to the wetlands, and islands or bodies of marine water deeper than six meters at low tide lying within the Wetlands.

3. USES AND IMPORTANCE OF WETLANDS

Wetlands play an important role in numerous ways and play a significant and important role towards mankind. The importance of Wetlands and its uses are as under:-

i. Controlling Flood

Wetlands play an important role in preventing and controlling flood. Major river system across the globe e.g. the Nile River, the Nijer River Delta, Mississippi River, Amazon River etc. produce large flood plains, which enable excess water to spread out over a wide area. Wetlands close to the Rivers often slows down the flow of sudden excess rain water or water on account of melting of snow in such a manner that it prevents sudden flooding of the streams and hence, is helpful in preventing sudden surge of water in the form of flash floods. Apart from this, wetland also acts as agent of storm protection and shoreline stabilization.

ii. Ground Water Replenishment and Purification

Wetlands systems are directly connected to the ground water and hence, play an important role in regulating both the quantity and quality of ground water. Wetlands are made of porous elements like lime stones and are often found in areas with highly fluctuating water table, hence, they play an important role in re-charging the ground water. The Wetlands system allows water to filter down through the soil and overlying rocks and hence, it acts as a mechanism for purification of water. Karst caves are the best examples of this system.

iii. Biodiversity

Wetlands systems have a rich biodiversity having a large number of species. Wetlands systems are a rich source of aquatic animals including fresh water fish, and diverse and distinct varieties of trees.

iv. Wetland Products

As Wetlands naturally produce a variety of vegetations and other ecological products, which can be utilized for personal and commercial use, hence, a variety of products such as fuel wood, salt, animal fodder, traditional medicines, fibers for textiles and dyes are some of the products, which are directly derived from the mangroves found in the Wetlands.

v. Climate Change

Wetlands have an important function in relation to climatic changes. The Wetlands system has unique ability to reduce carbon. The Wetlands have mitigation effect through their abilities to absorb carbon and are also a significant source of methane emissions. The Wetlands Systems are producers of a major green house gas namely Nitrous Oxide. Hence, the Wetland systems are having a major contribution towards climate.

4. PROTECTION OF WETLAND

Conversion of Wetland for commercial purposes e.g. mining, development, construction, grazing, agriculture, dumping, diverting flow of water, de-vegetation are some of the major threats to the Wetland systems across the world. Rapid industrialization and commercial development coupled

with un-scientific use of fresh water towards industries and other developmental activities have posed considerable threats to the Wetland systems. The uncontrolled exploitation of ground water has posed a major threat to the ability of nature to replenish ground water, as a result of which, the water tables have fallen to drastic levels across the world. Huge chunks of Wetlands are being gradually brought under agriculture, which has resulted into reduction of area under the Wetlands. Pollution through air, water and land and mindless discharge of pollutants has also affected the Wetland Systems across the world and have caused a serious threats to the biological diversities found in the Wetland.

Globally, Ramsar Convention on Wetlands was the first consolidated effort towards conservation and protection of Wetlands.

i. Ramsar Convention on Wetlands, 1971

Ramsar Convention is the oldest of the modern global inter governmental environmental agreements. Though, the negotiations on the treaty continued through 1960, but it was signed only in 1971. The Ramsar Convention on Wetlands was signed in 1971 in the City of Ramsar, Iran. The treaty was made to provide a regulatory frame work for the conservation and wise use of Wetland and its products.

The Ramsar convention identified Wetlands of international importance across the world for the purpose of enacting a regulatory frame work for the conservation and wise use of Wetland and its products. As per the information available on the website of Ramsar, as on March 2016, 2266 Wetland sites have been included in the list of Wetlands of international importance. United Kingdom have the maximum number of sites which are included in the list, whereas area wise Wetlands in Bolivia are the largest in the list.

Every three years, the representatives of the signatories to the treaty meet for the purpose of policy making and decisions, which are necessary for conservation and protection of Wetlands.

ii. Wetland Conservation in India

India is also a signatory to the Ramsar Convention on Wetlands and the Convention of Biological Diversity. Apart from government regulation, development of better monitoring methods is

needed to increase the knowledge of the physical and biological characteristics of each wetland resources, and to gain, from this knowledge, a better understanding of wetland dynamics and their controlling processes. India being one of the mega diverse nations of the world should strive to conserve the ecological character of these ecosystems along with the biodiversity of the flora and fauna associated with these ecosystems.

Wetlands are not delineated under any specific administrative jurisdiction. The primary responsibility for the management of these ecosystems is in the hands of the Ministry of Environment and Forests. Although some wetlands are protected after the formulation of the Wildlife Protection Act, the others are in grave danger of extinction. Effective coordination between the different ministries, energy, industry, fisheries revenue, agriculture, transport and water resources, is essential for the protection of these ecosystems. Protection laws and government initiatives with regards to Wetlands conservation in India is influenced by an array of policy and legislative measures. Some of the key legislations are given below:

1. Indian Fisheries Act, 1857
2. The Indian Forest Act, 1927
3. The Wildlife (Protection) Act, 1972
4. The Water (Prevention & Control of Pollution) Act, 1974
5. The Air (Prevention & Control of Pollution) Act, 1974
6. Territorial Water, Continental Shelf, Exclusive Economic Zone and other Marine Zones Act, 1976
7. Maritime Zone of India (Regulation and fishing by foreign vessels) Act, 1980
8. Forest (Conservation) Act, 1980
9. Environmental Protection Act, 1986 and various notifications framed under the act.
10. Coastal Zone Regulation Notification , 1991
11. National Conservation Strategy and Policy
12. Statement on Environment and Development 1992
13. National Policy And Macro level Action Strategy on Biodiversity,1999

The Ministry of Environment and Forest, Government of India has enacted elaborate programs on Wetland conservation and management in India. At present, India has around 26 Ramsar Recognized Wetlands and six more sites have been identified for the purpose of inclusion in the list of Ramsar Recognized Wetlands.

Recently, the Government of India has notified Wetlands (Conservation and Management) Rules, 2017 for the purpose of conservation and management of Wetland.

iii. Protection and Management of Wetlands

i. Making of wetlands strategy at national level

In order to protect and conserve wetlands, it is imperative that there must be strategy/planning in place at national level in all the countries across the globe, whether or not, they are signatories to the Ramsar Convention. The wetlands strategy must include three important aspects which are :

- (a). Conservation and management;
- (b). Prevention of loss and its restoration;
- (c). Sustainable management;

Apart from this, the other factors, which are important for the purpose of protection of Wetlands are explained as under:-

ii. Enacting effective legislations and its strict enforcement for the purpose of the protection of the wetland

In order to ensure protection of Wetlands, it is necessary that effective legislations must be enacted and enforced strictly. The United States of America has enacted Federal Legislations as well as State Legislations, which deals with the Wetland Protection and Conservation in U. S. A. The notable provisions are found in the Clean Water Act and numerous programs and policies made for the conservation and management of Wetlands in U. S. A. Similarly, in India also, National Wetlands Conservation and Management Program is in place, which is reviewed periodically. The legal frame work in India is found in numerous legislations, though there is no separate legislation

with regards to Wetland Conservation, but Wetlands conservation provisions are found in numerous legislations.

iii. Planning, Managing and Monitoring

The primary necessity of today is to protect the existing Wetlands, which are identified and included in the list of Recognized Wetlands and those Wetlands, which are important, but have not been included in the list of Recognized Wetlands. Hence, it is important that Wetlands that come under the protected area network have management plans but others do not. It is important for various stakeholders along with the local community and corporate sector to come together for an effective management plan. Active monitoring of these wetland systems over a period of time is essential.

iv. Research

There is a necessity for research in the formulation of national strategy to understand the dynamics of these ecosystems. This could be useful for the planners to formulate strategies for the mitigation of pollution. The scientific knowledge will help the planners in understanding the economic values and benefits, which in turn will help in setting priorities and focusing the planning process.

v. Comprehensive Inventory

Survey and preparation of an elaborate inventory is imperative for the purpose of protection and conservation of Wetlands. Hence, there must be an elaborate survey and preparation of inventory, which would take into consideration the various factors affecting the Wetlands. The survey and preparation of inventory can be done with the remote sensing and elaborate mappings through revenue records assessment and land use patterns. There has been no comprehensive inventory of all the Indian wetlands despite the efforts by the Ministry of Environment and Forests, Asian Wetland Bureau and World Wide Fund for Nature. The inventory should involve the flora, fauna, and biodiversity along with values. It should take into account the various stakeholders in the community too. In India some of the notable surveys of Wetlands, which have been undertaken are as under:-

- i. Asian Wetlands Directory, 1989;
- ii. Wetlands Directory published by the Ministry of Environment and Forest in 1990;
- iii. Identification of new Wetlands (Natural and Manmade);
- iv. Survey conducted by WWF India and Ministry of Environment and Forest in 1993;
- v. National Wetlands Information System and Updation of Wetland Inventory, a project sanctioned by Ministry of Environment and Forest

These are some of the few notable surveys undertaken.

vi. Capacity Building:

Capacity building is a major tool, without which no conservation activity is possible. Hence, there must be a good infrastructure in place. Apart from this, it is very necessary that there must be experts and well-trained people, who are engaged directly for the purpose of Wetlands conservation and protection. The capacity building can also be strengthened by including training to policy makers, senior officials, organizations, stake holders and others. The Government of India in its 10th Five Years Plan have included several training programs.

vii. Coordinated approach

Since wetlands are common property with multi-purpose utility, their protection and management also need to be a common responsibility. An appropriate forum for resolving the conflict on wetland issues has to be set up. It is important for the ministries to allocate sufficient funds towards the conservation of these ecosystems.

ix. Community Participation

No process of conservation, protection and management is possible without the active participation of the local population, whose livelihood are directly dependent upon the resources of the Wetland. There are areas near the Wetlands, where local people have been using the Wetlands since long and is a major source of livelihood. Hence, it is imperative that in order to ensure conservation, protection and management of Wetlands community participation must be encouraged, which can be achieved by carrying on an elaborate assessment of the existing pattern

of usage of wetlands in that particular area and using techniques to blend both traditional and latest scientific technologies to achieve long term conservations go by making people aware of the importance of protection and conservation of Wetlands. However, this can only be achieved, if the local community is encouraged to protect Wetlands and reduce its exploitation by ensuring that the Government and Stake Holders provide elaborate alternative arrangements to the local population, so that they are not deprived of their basic necessities.

5. CONCLUSION

For achieving any sustainable success in the protection of these wetlands, awareness among the general public, educational and corporate institutions must be created. The policy makers, at various levels along with site managers need to be educated. As the country's wetlands are shared, the bi-lateral cooperation in the resource management needs to be enhanced.

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